

Notes: Bolton and Scharfstein (AER, 1990)

A Theory of Predation Based on Agency Problems in Financial Contracting

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1 Big picture

- **Question.** Why can a cash-rich rival drive out a financially constrained competitor, and why do the competitor's creditors not simply relax financing constraints once predation becomes a threat?
- **Core answer.** The financing constraints are not imposed from outside. They arise endogenously because investors need the threat of termination to keep managers from diverting resources away from investors.
- **Main mechanism.** The constrained firm's manager can hide or divert resources when profits are not contractible. Investors therefore commit to terminate funding after poor performance. A rival can exploit this by taking actions that make poor performance more likely, increasing the chance that the funding cutoff becomes binding.
- **Why this is a theory of predation.** The predator does not need to convince the target that staying in the market is intrinsically unprofitable. Everyone knows the target's expected project value is positive. Predation works by worsening the target's agency problem with its financiers.
- **How the paper differs from signaling stories.** Profits are assumed to be independent across periods in the baseline model, so predation is *not* about changing beliefs about future demand or costs. It is about changing the chance that an inefficient liquidation rule is triggered.
- **Baseline contracting insight.** In a one-period relationship, outside finance is impossible because the firm would always report the low-profit state and pay only the minimum verifiable amount. In two periods, the threat of no refinancing creates discipline and makes financing possible.
- **Predation and finance interact.** The contract that best solves the incentive problem is also the one that makes predation most attractive. The more sharply refinancing depends on performance, the bigger the gain to a rival from pushing performance down.
- **Observable-contract case.** If the predator can see the financing contract, the investor can try to deter predation by making refinancing less sensitive to performance.
- **Surprising main result.** When low profits imply a negative continuation value for investors ($\pi_1 < F$), the optimal deterrence strategy is often a *shallow-pocket* strategy: lower the probability of refinancing even after good performance, rather than promise refinancing after bad performance.
- **Why the shallow-pocket result is surprising.** One might think the natural anti-predation response is to give the target a deeper pocket by promising more support in bad states. But doing that is costly for investors because refinancing low-profit firms loses money. Lowering continuation after good performance weakens the predator's incentive at lower contracting cost.
- **Unobservable-contract case.** If the rival cannot observe the contract, the constrained firm cannot credibly precommit to the softer anti-predation contract. Then the investor chooses the steep incentive contract ex post, and predation occurs in equilibrium.
- **Broader contribution.** Financial structure affects not just financing costs, but also product-market outcomes. "Deep pockets" matter because they change how vulnerable a firm is to strategic attacks that operate through capital-market frictions.

- **Takeaway.** Predation can be rational even when the target’s projects remain positive-NPV. Agency problems in financial contracting create fragile continuation rules, and rivals can strategically manipulate the likelihood that those rules force exit.

2 Data and measurement (key objects)

2.1 This is a theory paper: the key objects are model primitives

There is no empirical dataset, no regression design, and no statistical identification exercise. The “measurement” objects are the firms, the financing contract, the profit states, and the predation technology.

2.2 Two firms and two periods

The economy has two firms, labeled A and B , that compete in periods 1 and 2.

- Firm A has a **deep pocket**: it can finance the fixed operating cost internally.
- Firm B has a **shallow pocket**: it must raise outside funds from investors.

At the beginning of each period, operating requires paying a fixed cost F .

2.3 Firm B’s profit process

Firm B ’s gross operating profit before financing costs is either low or high:

$$\pi \in \{\pi_1, \pi_2\}, \quad \pi_1 < \pi_2. \quad (1)$$

At the start of each period, all players believe that profit is low with probability θ . In the baseline model, profits are independently distributed across periods.

2.4 Expected profit

Define expected per-period profit as

$$\bar{\pi} = \theta\pi_1 + (1 - \theta)\pi_2. \quad (2)$$

The key maintained assumptions are

$$\pi_1 < F < \bar{\pi}. \quad (3)$$

Interpretation.

- Low current profit makes second-period financing privately unattractive to investors.
- Average project value is still positive, so liquidation after a bad first period is *ex post* inefficient.

2.5 Profit noncontractibility

The investor cannot write a fully enforceable contract contingent on realized profits. The paper offers two interpretations:

- profits are privately observed by the firm, or
- profits are observable to both sides but not verifiable in court.

The investor can, however, force a minimum payment of π_1 each period. This is the verifiable component of profits.

2.6 First-period transfer schedule

If the firm reports first-period profit π_i , the transfer to the investor at date 1 is

$$R_i, \quad i = 1, 2. \quad (4)$$

2.7 Refinancing decision

If the firm reports π_i in period 1, the investor refinances second-period production with probability

$$\beta_i \in [0, 1], \quad i = 1, 2. \quad (5)$$

The paper also notes that β_i can be reinterpreted as a *capacity-expansion* parameter rather than a literal randomization device.

2.8 Second-period transfer schedule

Let

$$R^i \quad (6)$$

be the second-period transfer conditional on the first-period report being π_i .

Interpretation. Because the second-period transfer cannot be made contingent on second-period realized profit, it depends only on the first-period report.

2.9 Predation technology

Predation is modeled as a costly action by firm A .

- By paying cost $c > 0$, firm A increases the probability that firm B earns low first-period profit from θ to μ , where $\mu > \theta$.
- If firm B exits, firm A becomes a monopolist in period 2 and earns expected profit π^m .
- If firm B stays, firm A remains in duopoly and earns expected profit π^d .

Define the normalized predation-cost parameter

$$\Delta \equiv \frac{c}{(\mu - \theta)(\pi^m - \pi^d)}. \quad (7)$$

Interpretation. Smaller Δ means predation is cheaper relative to the gain from eliminating the rival.

2.10 Contract observability

A key institutional distinction in the paper is whether firm A can observe the contract chosen by firm B and its investor.

- If the contract is observable, financing policy can strategically influence the rival.
- If the contract is unobservable, the rival must conjecture the contract and the target cannot precommit to an anti-predation financing policy.

3 Models and empirical specifications (all equations and interpretation)

3.1 Why one-period finance fails

If there were only one period, the investor would never fund the project. The firm would always report the low-profit state and pay at most π_1 , so the investor's payoff would be below the fixed cost F whenever $\pi_1 < F$.

Economic message. Outside finance only becomes possible because a two-period relationship creates a continuation threat.

3.2 The baseline contracting problem

At date 0, a single investor offers firm B a take-it-or-leave-it contract. The investor chooses transfers and continuation probabilities to maximize expected payoff:

$$\max_{\{\beta_i, R_i, R^i\}} -F + \theta[R_1 + \beta_1(R^1 - F)] + (1 - \theta)[R_2 + \beta_2(R^2 - F)]. \quad (8)$$

The contract is subject to incentive compatibility, limited liability, and participation.

3.3 Incentive compatibility

The relevant incentive constraint is that a high-profit firm should not pretend to be low-profit:

$$\pi_2 - R_2 + \beta_2(\bar{\pi} - R^2) \geq \pi_2 - R_1 + \beta_1(\bar{\pi} - R^1). \quad (1)$$

Interpretation.

- Reporting low profit reduces the first-period payment.
- But it also lowers the chance of receiving continuation surplus in period 2.
- The entire discipline device is the threat of losing that continuation value.

3.4 Limited liability

The first-period transfer cannot exceed realized profit:

$$\pi_i \geq R_i, \quad i = 1, 2. \quad (2)$$

The second-period transfer cannot exceed available cash after the first-period payment plus the minimum verifiable second-period profit:

$$\pi_i - R_i + \pi_1 \geq R^i, \quad i = 1, 2. \quad (2')$$

3.5 Participation

Firm B accepts the contract only if its expected value is nonnegative:

$$\theta[\pi_1 - R_1 + \beta_1(\bar{\pi} - R^1)] + (1 - \theta)[\pi_2 - R_2 + \beta_2(\bar{\pi} - R^2)] \geq 0. \quad (3)$$

3.6 Two lemmas that collapse the problem

The paper proves two simplifying results.

- **Lemma 1.** The incentive-compatibility constraint (1) binds at the optimum.
- **Lemma 2.** There exists an optimal contract with equal second-period transfers: $R^1 = R^2 = \pi_1$.

Interpretation. Once the firm reaches period 2, the investor no longer has a continuation threat, so he can extract at most the minimum verifiable amount π_1 .

3.7 The implied high-state payment

Given the binding incentive constraint, the first-period payment in the high-profit state satisfies

$$R_2 = \pi_1 + (1 - \beta_1)(\bar{\pi} - \pi_1). \quad (5)$$

Interpretation.

- The firm earns continuation surplus $\bar{\pi} - \pi_1$ if it stays alive.
- Lowering β_1 makes lying more costly because a high-profit firm that reports low loses more continuation value.
- The investor can then demand a bigger payment in the high-profit state.

3.8 Reduced-form objective

Using the lemmas, the optimization simplifies to

$$\max -F + R_1 + \beta_2(1 - \theta)(\bar{\pi} - F) - \beta_1[\theta F + (1 - \theta)\bar{\pi} - \pi_1]. \quad (4)$$

Interpretation.

- Raising β_2 is valuable because the investor refinances only after a good first-period report.
- Raising β_1 is costly because it forces refinancing after a bad report, precisely when continuation is privately unattractive.

3.9 Proposition 1: optimal contract without predation

The solution is

$$R_1^* = \pi_1, \quad \beta_1^* = 0, \quad R_2^* = \bar{\pi}, \quad \beta_2^* = 1, \quad R^{1*} = R^{2*} = \pi_1. \quad (9)$$

The investor funds the firm at date 0 if and only if

$$F < \bar{\pi} - \frac{\bar{\pi} - \pi_1}{2 - \theta}. \quad (10)$$

Interpretation.

- If first-period profit is low, the investor takes π_1 and terminates the firm.
- If first-period profit is high, the investor takes $\bar{\pi}$ and refinances for period 2.
- The contract is intentionally harsh in bad states because that harshness creates the truth-telling incentive that makes external finance feasible.

3.10 Why liquidation is credible

Although liquidation after π_1 is inefficient from a total-surplus perspective, it is credible from the investor's perspective because

$$\pi_1 - F < 0. \quad (11)$$

So once the low-profit state is realized, refinancing loses money for the investor.

3.11 Predation technology and the benefit to the rival

Suppose firm A can spend c to raise the probability that firm B draws low first-period profit from θ to $\mu > \theta$. Given contract (β_1, β_2) , predation changes the exit probability by $\beta_2 - \beta_1$. Hence the benefit to firm A from predation is

$$(\beta_2 - \beta_1)(\mu - \theta)(\pi^m - \pi^d). \quad (12)$$

Firm A preys if and only if

$$(\beta_2 - \beta_1)(\mu - \theta)(\pi^m - \pi^d) > c, \quad (13)$$

which is equivalent to

$$\beta_2 - \beta_1 > \Delta. \quad (14)$$

Interpretation. Predation is more attractive when the financing contract makes survival sharply state-contingent.

3.12 Observable contracts and the no-predation constraint

If firm A can observe the contract, the investor can choose β_1 and β_2 to satisfy

$$(\beta_2 - \beta_1)(\mu - \theta)(\pi^m - \pi^d) \leq c, \quad (6)$$

that is,

$$\beta_2 - \beta_1 \leq \Delta. \quad (6')$$

3.13 Proposition 2: optimal predation deterrence

When deterrence is chosen, the optimal contract is

$$\beta_1^* = 0, \quad \beta_2^* = \Delta, \quad R_1^* = \pi_1, \quad R_2^* = \Delta\bar{\pi} + (1 - \Delta)\pi_1 < \bar{\pi}. \quad (15)$$

The associated expected profit is

$$\pi_1 - F + (1 - \theta)\Delta(\bar{\pi} - F). \quad (16)$$

Firm B enters if and only if

$$\pi_1 - F + \max\{(1 - \theta)\Delta, 1 - \mu\}(\bar{\pi} - F) \geq 0. \quad (17)$$

Conditional on entering, firm B chooses to deter predation when

$$(1 - \theta)\Delta \geq 1 - \mu. \quad (18)$$

Why the result is “shallow pocket.”

- Reducing β_2 and increasing β_1 both shrink $\beta_2 - \beta_1$ and therefore both weaken the predator's incentive.
- But increasing β_1 is expensive because the investor then refinances even after a low-profit report, when continuation loses money.
- Lowering β_2 achieves the same deterrence with lower incentive cost to the investor.

3.14 Why a public randomization device matters

The observable-contract deterrence result relies on allowing β_i to lie in the interval $[0, 1]$. Without a public randomization device, the feasible deterministic values are 0 or 1. Then the only feasible entry contract has $(\beta_1, \beta_2) = (0, 1)$ and predation cannot be deterred in this simple binary environment.

3.15 What changes when the low state is profitable

The shallow-pocket result is driven by the assumption that continuation after low profit is privately unprofitable for investors. If instead $\pi_1 > F$, refinancing after the low state no longer loses money. Then increasing β_1 becomes more attractive and a deep-pocket response can dominate.

Interpretation. The most distinctive result of the paper is tightly linked to the case where bad current performance makes external continuation privately unattractive.

3.16 Renegotiation

If profits are observable to the firm and investor, even if not verifiable, then the original cutoff contract may not be renegotiation-proof. If the firm reports low profit while actually having high profit, it may retain enough internal cash to finance period 2 itself or to renegotiate with the investor.

If the firm captures a fraction α of the efficiency gain from continuing, a renegotiation-proof high-state payment becomes

$$\bar{\pi} - \alpha(\bar{\pi} - F). \quad (19)$$

Interpretation. The more bargaining power the firm has in renegotiation, the less rent the investor can extract up front, and the harder it is to support financing at date 0.

3.17 Positive serial correlation in profits

The paper also studies the case where first-period and second-period profits are positively correlated. Let $\mathbb{E}(\pi \mid \pi_i)$ be expected second-period profit conditional on first-period profit π_i . Then the optimal contract still sets

$$\beta_1^* = 0, \quad \beta_2^* = 1, \quad R_1^* = \pi_1, \quad R_2^* = \mathbb{E}(\pi \mid \pi_2). \quad (20)$$

Interpretation. Positive serial correlation strengthens the baseline result because losing refinancing is even more painful after a truly good first-period outcome, making misreporting less attractive.

3.18 Unobservable contracts

If firm A cannot observe the contract, firm B and the investor cannot credibly precommit to the anti-predation contract. In the binary model with $\pi_1 < F$, the investor's dominant strategy is simply to choose the privately optimal incentive contract

$$(\beta_1^*, \beta_2^*) = (0, 1). \quad (21)$$

Firm A then preys, so the unique Nash equilibrium (conditional on entry) has predation.

3.19 Proposition 3: entry with hidden contracts

With unobservable contracts, firm B enters if and only if

$$\pi_1 - F + (1 - \mu)(\bar{\pi} - F) > 0. \quad (22)$$

If it enters, the contract is the baseline Proposition 1 contract and firm A preys.

Interpretation. Hidden contracts destroy commitment. Even though a softer contract would deter predation, the investor does not want to choose it once the rival's behavior is taken as given.

4 Identification and credibility (what makes the model persuasive)

4.1 What “identification” means in this paper

Because this is a theory paper, credibility comes from the logic of the mechanism rather than from an econometric design. The question is whether the assumptions isolate a believable channel through which finance changes product-market outcomes.

4.2 Financial constraints are endogenous, not assumed

A major strength of the paper is that it does not simply assume the target is financially constrained. The financing friction is derived from a contracting problem with unverifiable profits. That makes the predation result more compelling: the rival exploits a constraint that is privately optimal for the investor-manager relationship.

4.3 Why the termination threat is credible

The argument needs a continuation threat that can actually be carried out. That is exactly what $\pi_1 < F$ delivers.

- After a bad first-period outcome, the investor does not want to refinance.
- So the threat to terminate is not a cheap talk threat or an empty penalty.
- This distinguishes the model from inspection-based debt models in which the ex post penalty may itself be noncredible.

4.4 Why this is not a belief-manipulation model

The model deliberately assumes profits are independent across periods in the baseline. That shuts down the usual limit-pricing logic in which predation changes the target's beliefs about future profitability.

Interpretation. Predation here works through capital-market fragility, not through misleading the rival about market fundamentals.

4.5 The observability assumption is central

Whether contracts are observable to the predator determines whether the constrained firm can precommit to a softer anti-predation policy.

- If contracts are observable, the investor can strategically choose $\beta_2 - \beta_1$.
- If contracts are hidden, the investor reverts to the privately optimal steep contract and predation reappears.

Interpretation. The paper therefore links disclosure and transparency to strategic market outcomes in a subtle way.

4.6 Why the shallow-pocket result is not universal

The paper is careful about its own limits.

- If $\pi_1 > F$, continuation after bad performance is no longer privately loss-making, so deep-pocket support can become optimal.
- If renegotiation is frictionless and efficient, cutoff contracts are harder to sustain.
- If one insists on deterministic funding rules, the anti-predation randomized contract may be infeasible in the binary model.

Interpretation. The striking policy lesson is not “always make firms shallower pocketed.” It is that the optimal anti-predation response depends on the private cost of continuation in bad states.

4.7 Why the model feels institutionally relevant

The paper connects naturally to real-world financing arrangements:

- venture-capital staged financing,
- debt obligations and payout commitments,
- dividend or repurchase decisions that force firms back to the capital market,
- and the general idea that internal liquidity protects firms from external-finance shocks.

Interpretation. The model is abstract, but the institutions it is trying to capture are familiar and economically important.

4.8 Relation to later literature

The paper also anticipates later work emphasizing that capital structure can influence competitive behavior not only by changing risk-taking incentives, but by changing the entire continuation value of staying in the market.

Bottom line. The mechanism is persuasive because it is built from a small number of transparent assumptions, each of which has a clear institutional interpretation.

5 Tables and figures

This is a **theory paper** with no regression tables and no empirical figures. The key things to “read” are the contract formulas, the three propositions, and the discussion of when different anti-predation responses are optimal.

5.1 Proposition 1: the baseline financing contract

Read:

- After a low first-period report, the investor takes π_1 and terminates funding.
- After a high first-period report, the investor takes $\bar{\pi}$ and refinances with probability one.
- Second-period transfers collapse to the minimum verifiable amount π_1 .

Economic message. The threat of termination is not a side detail. It is the core contracting device that makes outside finance feasible.

5.2 Equation (5): how the continuation threat creates rent extraction

Read:

- The high-state first-period payment rises when β_1 falls.
- The reason is that a lying high-profit firm loses more continuation surplus if it reports the low state.
- The difference $\bar{\pi} - \pi_1$ is the rent created by future operation.

Economic message. This equation is the heart of the agency argument. It shows exactly why harsh continuation rules can relax current incentive constraints.

5.3 Constraint (6’): the no-predation boundary

Read:

- Firm A cares only about how much predation changes the target’s continuation probability.
- That change is measured by $\beta_2 - \beta_1$.
- Deterring predation therefore means compressing the difference between continuation after good and bad performance.

Economic message. The product market “reads” the financing contract through one statistic: how performance-sensitive survival is.

5.4 Proposition 2: the shallow-pocket deterrence result

Read:

- The optimal deterrence contract sets $\beta_1 = 0$ and $\beta_2 = \Delta$.
- So the investor lowers refinancing after good performance rather than promising extra refinancing after bad performance.
- Entry and deterrence depend on comparing $(1 - \theta)\Delta$ with $1 - \mu$.

Economic message. The paper's most novel insight is that a firm may optimally defend itself against predation not by becoming tougher in bad states, but by becoming less rewarding to attack in the first place.

5.5 The discussion when the low state is profitable

Read:

- If the investor can still make money after a low-profit state, increasing β_1 becomes attractive.
- In that case, a deep-pocket response can dominate.
- The shallow-pocket result therefore depends on bad states being genuinely painful for outside financiers.

Economic message. The paper's comparative statics are economically sharp: the sign of $\pi_1 - F$ changes the nature of the optimal defense.

5.6 Proposition 3: hidden contracts restore predation

Read:

- If contracts are unobservable, the investor chooses the privately optimal steep contract $(0, 1)$.
- Firm A then preys in equilibrium.
- Entry is weaker than in the observable-contract case because firm B cannot commit to the softer deterrence contract.

Economic message. Strategic commitment requires visibility. If rivals cannot see the financing policy, the target cannot use it to shape rival behavior.

5.7 Section I.B: renegotiation and extensions

Read:

- Efficient renegotiation can undermine the force of the cutoff rule.
- Positive serial correlation in profits strengthens the truth-telling role of the continuation threat.
- The β_i 's can be interpreted as staged capital expansion rather than literal randomization.

Economic message. The central mechanism is robust, but the exact contract form depends on bargaining, observability, and the detailed source of continuation value.

6 Short summary

Bolton and Scharfstein build a two-period model in which financial constraints arise endogenously because investors need the threat of no refinancing to discipline managers when profits are not contractible. A rival can prey on the constrained firm by making poor first-period performance more likely, thereby increasing the probability that the financing cutoff triggers inefficient exit. If the rival can observe the contract, the investor may optimally deter predation by reducing the sensitivity of refinancing to performance. With $\pi_1 < F$, the surprising optimal response is often a shallow-pocket strategy: lower refinancing after good performance rather than promise support after bad performance. If the contract is unobservable, however, the firm cannot credibly precommit to that softer policy, so the investor reverts to the steep incentive contract and predation occurs in equilibrium.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that rational predation can arise from agency problems in financial contracting. A target firm may exit not because continued production is fundamentally unprofitable, but because its optimal outside-finance contract makes continuation fragile after bad performance.

7.2 Economic intuition

Investors want a hard refinancing cutoff because it helps keep managers honest. Rivals want to make that cutoff bind. Once those two facts are combined, product-market competition and capital-market contracting become inseparable. Predation works by shifting the target into the region where the investor's privately optimal rule is to stop funding.

7.3 Takeaway

Bolton and Scharfstein's lasting insight is that *financial vulnerability itself can be a strategic state variable*. Deep pockets matter not only because they let firms spend more, but because they shield firms from rivals that try to exploit the target's agency-based financing fragility.